

Realization of a Chebyshev-Like Radiation Pattern in a Plural Circular-Array Antenna Using Quadratic Excitation Distribution

学部 4 年 1082 番 山田 一輝

1. Abstract

This paper demonstrates the realization of a Chebyshev-like radiation pattern, characterized by constant and low sidelobe levels, using a plural circular-array antenna. While this antenna offers excellent control over the radiation pattern, it often suffers from high manufacturing costs due to many elements. In this study, we apply a quadratic excitation distribution with a -30dB edge level to the array elements. The resulting radiation pattern is shown to exhibit sidelobes of approximately -29dB, closely mimicking a Chebyshev response. We perform a comparative analysis against a linear Dolph-Chebyshev array, revealing that applying its coefficients directly to the circular array results in low but uneven sidelobes. The study clarifies the relationship and trade-offs between linear and plural circular-array configurations for achieving desirable, low-sidelobe patterns.

2. Antenna Design and Methodology

The antenna under analysis is a 4-ring plural circular-array antenna. The radiation pattern is calculated using the following general expression:

$$FA(\theta, \varphi) = \sum_{m=0}^M \sum_{n=1}^N A_{mn} \exp\{j k r_m \sin \theta \cos(\varphi - \Psi_n)\}$$

where

A_{mn} is the excitation of the element (m, n) . The radiator elements are assumed to be isotropic.

To achieve the Chebyshev-like pattern, the excitation coefficients A_{mn} are determined from a continuous quadratic field distribution with a -30dB edge level, given by:

$$f(r) = 1 - a(r/a)^2$$

where

r is the radial location and a is the radius of the aperture. For comparison, a 9-element Dolph-Chebyshev linear array is also studied, and its coefficients are applied directly to the circular arrays.

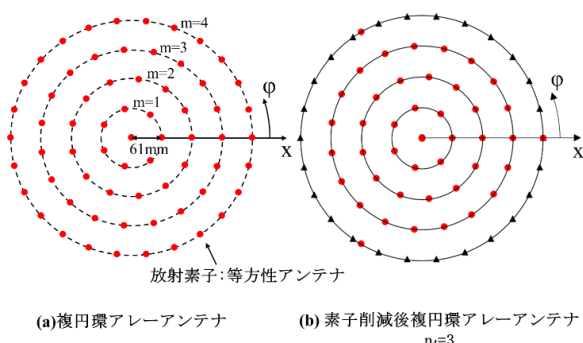


Figure 1: Antenna Configuration

3. Results and Discussion

2.1. Pattern Synthesis with Quadratic Distribution

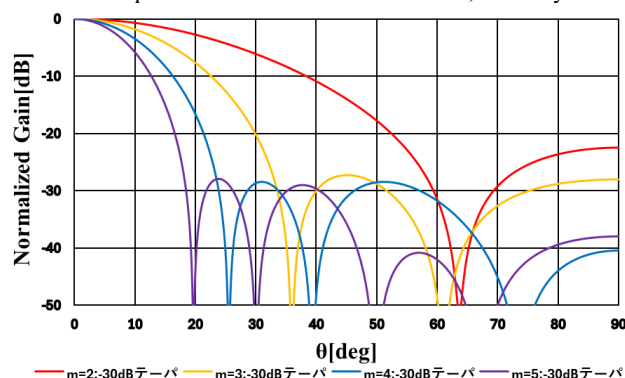
The 4-ring circular array with the quadratic coefficient distribution produced a radiation pattern with constant and low sidelobes of

approximately -29dB. This can be described as a "Chebyshev-like" pattern. In contrast, the same array with a uniform excitation exhibited a first sidelobe of -17dB, consistent with theoretical values for a uniformly illuminated aperture. While the main beam of the quadratic distribution case is wider, the sidelobe performance is significantly superior.

Figure 2: Radiation Pattern with Quadratic Distribution

2.2. Impact of Element Reduction for Cost Savings

To evaluate a practical method for cost reduction, we analyzed the



impact of reducing the number of elements (n_4) in the outermost ring of the 4-ring array with a -30dB quadratic taper applied. The results showed that the levels of the first and second sidelobes remained remarkably stable and nearly equal as the number of elements was reduced from 26 down to 9. Below this count, a discrepancy of up to 1.55 dB emerged between these two sidelobes. In contrast, the third sidelobe's behavior was less predictable, becoming highly erratic with fewer than 15 elements, even though it showed a gradual increase in level down to that point. This analysis indicates that applying a tapered distribution provides significant robustness against element reduction in the outer ring, making it a viable strategy for minimizing cost while maintaining acceptable sidelobe performance, particularly for the first and second sidelobes.

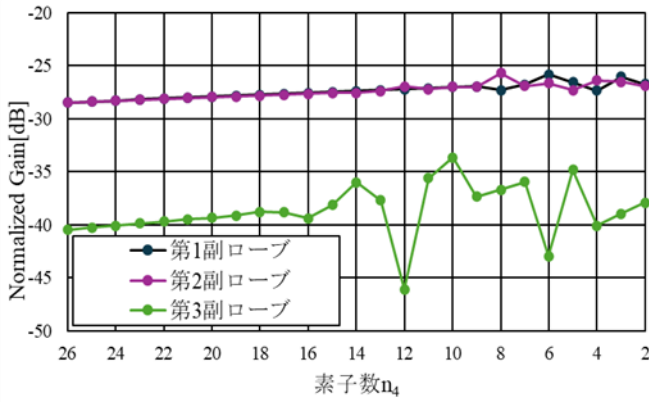


Figure 3. Sidelobe level (SLL) variation as a function of the number of elements in the outermost ring (n_4) for the 4-ring array with a -30dB quadratic taper.

2.3. Application of Dolph-Chebyshev Coefficients

Next, the excitation coefficients from a -30dB Dolph-Chebyshev linear array were applied to the four circular arrays to study the relationship between the two antenna types. The Dolph-Chebyshev linear array itself produced its designed pattern, exhibiting constant -30dB sidelobes as expected. However, when these same coefficients were applied to the plural circular-array, the resulting radiation pattern showed low sidelobes with the highest being -29dB, but they were not constant in either level or width. A key observation is that the excitation coefficients required for the Chebyshev distribution are much higher at the edge of the aperture compared to those of the quadratic distribution. This indicates that the quadratic distribution achieves its low sidelobes through a significantly stronger taper.

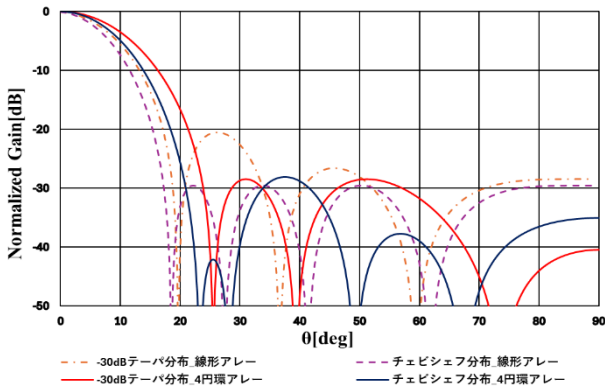


Figure 4. Radiation pattern of a 4-ring circular array using Dolph-Chebyshev coefficients (solid line), compared with the pattern of the original Dolph-Chebyshev linear array

4. Conclusion

This study successfully demonstrated that a Chebyshev-like radiation pattern with low, constant sidelobes can be realized in a plural circular-array antenna by adopting a quadratic excitation distribution. A comparison with a Dolph-Chebyshev array revealed a critical insight: a direct application of linear array coefficients to a plural circular-array does not preserve the constant sidelobe characteristic, although low sidelobe levels are maintained. This indicates the existence of a transformation relationship between linear and circular array geometries that warrants further investigation. The quadratic distribution method

provides a simple yet effective way to achieve excellent sidelobe performance, offering a practical design alternative to complex Chebyshev synthesis for circular apertures.

References

- [1] T. Takano and K. Saegusa, "Proposal of a Plural Circular Array Antenna to Control the Radiation Pattern", ISAP 2023, Kuala Lumpur, Malaysia, November, 2023.

まず、論文全体の構成を、単なる翻訳ではなく、国際的な学術論文の標準形式である「Abstract」「Methodology」「Results and Discussion」「Conclusion」に沿って再構築しました。タイトルは、単に「～の研究」とするのではなく、「Realization of... (～の実現)」という、研究成果を能動的かつ力強くアピールする表現に変更しました。同時に、Chebyshev-Like や Plural Circular-Array Antenna、Quadratic Excitation Distribution といった具体的な専門用語をタイトルに含めることで、その分野の専門家が瞬時に論文の価値を判断でき、学術データベースでの検索性も挙げられたと思います。

論文全体で使われる用語は、一度 IEEE で投稿した英語論文スタイルに合わせて Plural Circular-Array Antenna に統一し、より具体的な Quadratic Excitation Distribution といった専門性の高い表現へと言い換えました。Abstract では、論文の役割を客観的に示す This paper demonstrates という定番の現在形の表現で始め、研究内で行った具体的な操作は we applied のように過去形で記述しました。また、characterized by (～によって特徴づけられる) や、既存技術の課題を示す suffers from といった語彙を選んでいきます。

結果と考察のセクションでは、In contrast, (対照的に) や However, (しかしながら) といった接続詞を効果的に用いることで、論理的で自然な流れを持つパラグラフへと修正しました。これにより、二次関数分布と一様分布の比較点が明確になったと思います。結果の記述には、produced a radiation pattern のように過去形を用い、-29dB といった具体的な数値データを加えることで客観的な信頼性を高めています。さらに、consistent with theoretical values (理論値と一致) という表現で結果の妥当性を補強し、remarkably stable (著しく安定している) や significantly superior (著しく優れている) といった副詞や形容詞で、データが示す事実の重要性を強調しました。

手法の記述では、The antenna under analysis is... のように受動態を用いることで、主観を排した客観性を大事にしました。そして結論部分では、This study successfully demonstrated (本研究は～を見事に論証した) や revealed a critical insight (決定的に重要な知見を明らかにした) といった力強い表現で研究の貢献を要約しました。最後に、今後の研究の可能性を示唆する warrants further investigation (さらなる調査に値する) や、研究の実用的な価値を示す offering a practical design alternative (実用的な設計上の代替案を提示する) といった学術論文特有の表現で締めくくりました。

最も使用した AI は Gemini であり、理由としては使用制限がないのはもちろんですが、日本語論文からいらない文章や図、構成を考えるのがチャット GPT よりもすぐれていたからです。